

# Tutorial Sheet 4

## *Data-driven control*

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**Exercise 1.** Consider a dynamical system

$$x(k+1) = Ax(k) + Bu(k),$$

where  $x \in \mathbb{R}^n$  denotes the state and  $u \in \mathbb{R}^m$  denotes the control input. The matrices  $A$  and  $B$  are unknown, but we the full state (and control input) is available for measurements.

- a) Show how you can use data collected offline to construct data matrices  $U_{0,T-1}$ ,  $X_{0,T-1}$  and  $X_{1,T}$  that can be used to identify the matrices  $A$  and  $B$ . Make sure to define how the above data matrices are constructed and state any conditions that the data must satisfy.
- b) Show how you can use the data matrices to represent the open-loop system purely in terms of data, without relying on any system identification.
- c) Suppose now that we are interested in designing a feedback control law of the form

$$u(k) = Kx(k).$$

Let  $G_K \in \mathbb{R}^{T \times n}$  be a matrix satisfying

$$\begin{bmatrix} X_{0,T-1} \\ U_{0,T-1} \end{bmatrix} G_K = \begin{bmatrix} I \\ K \end{bmatrix}.$$

Suppose that

$$\text{rank} \begin{bmatrix} X_{0,T-1} \\ U_{0,T-1} \end{bmatrix} = n + m.$$

Derive an expression for the closed-loop system, *i.e.*

$$x(k+1) = (A + BK)x,$$

that depends on the available data and on the intermediate decision variable  $G_K$ .

- d) Considering the above data-representation of the closed-loop system, show how  $K$  should be designed given some appropriate selection of  $G_K$ .

$$(a) \quad U_{0,T-1} = [u_d(0), u_d(1), u_d(2), \dots, u_d(T-1)] \in \mathbb{R}^{m \times T}$$

$$X_{0,T-1} = [x_d(0), x_d(1), x_d(2), \dots, x_d(T-1)] \in \mathbb{R}^{n \times T}$$

$$X_{1,T} = [x_d(1), x_d(2), x_d(3), \dots, x_d(T)] \in \mathbb{R}^{n \times T}$$

$$\therefore x(k+1) = Ax(k) + Bu(k)$$

$$\therefore X_{1,T} = AX_{0,T-1} + BU_{0,T-1} = [A \ B] \begin{bmatrix} X_{0,T-1} \\ U_{0,T-1} \end{bmatrix}$$

$$\therefore [A \ B] = X_{1,T} \begin{bmatrix} X_{0,T-1} \\ U_{0,T-1} \end{bmatrix}^{\dagger} = X_{1,T} \begin{bmatrix} X_{0,T-1} \\ U_{0,T-1} \end{bmatrix}^T \left( \begin{bmatrix} X_{0,T-1} \\ U_{0,T-1} \end{bmatrix} \begin{bmatrix} X_{0,T-1} \\ U_{0,T-1} \end{bmatrix}^T \right)^{-1}$$

$$(b) \quad x(k+1) = [A \ B] \begin{bmatrix} x(k) \\ u(k) \end{bmatrix} = X_{1,T} \begin{bmatrix} X_{0,T-1} \\ U_{0,T-1} \end{bmatrix}^{\dagger} \begin{bmatrix} x(k) \\ u(k) \end{bmatrix}$$

(c) let  $G_k \in \mathbb{R}^{T \times n}$  be a solution of

$$\begin{bmatrix} X_{0,T-1} \\ U_{0,T-1} \end{bmatrix} G_k = \begin{bmatrix} I \\ K \end{bmatrix}$$

Therefore, let  $u(k) = Kx(k)$

$$x(k+1) = Ax(k) + Bu(k) = (A + BK)x(k) = [A \ B] \begin{bmatrix} I \\ K \end{bmatrix} x(k)$$

$$= [A \ B] \begin{bmatrix} X_{0,T-1} \\ U_{0,T-1} \end{bmatrix} G_k x(k)$$

$$= X_{1,T} G_k x(k)$$

(d) According to the Lyapunov equation, given any  $Q > 0$ , there is  $P = P^T$  such that

$$(A + BK)^T P (A + BK) - P + Q = 0$$

$$\therefore (A + BK)^T P (A + BK) - P < 0$$

$$\therefore (A + BK) P (A + BK)^T - P < 0$$

$$\therefore (X_{1,T} G_k) P (G_k^T X_{1,T}^T) - P < 0$$

Let  $Q = G_k P$ , therefore, exist  $P = P^T > 0$  that

$$\therefore \begin{cases} X_{0,T-1} Q = X_{0,T-1} G_K P = I P = P \\ U_{0,T-1} Q = U_{0,T-1} G_K P = K P \end{cases}$$

$$\therefore G_K = G_K P P^{-1} = Q P^{-1}$$

$$\therefore (X_{1,T} G_K) P (G_K^T X_{1,T}^T) - P = X_{1,T} Q (Q P^{-1})^T X_{1,T}^T - P = X_{1,T} Q P^{-1} (X_{1,T} Q)^T - P$$

$$\therefore \begin{cases} (X_{1,T} Q) P^{-1} (X_{1,T} Q)^T - P < 0 \\ P > 0 \end{cases}$$

satisfied the condition for positive definiteness via the Schur complement

$$M = \begin{bmatrix} \bar{A} & \bar{B} \\ \bar{B}^T & \bar{C} \end{bmatrix} > 0 \text{ if and only if } \begin{cases} \bar{A} - \bar{B} \bar{C}^{-1} \bar{B}^T > 0 \\ \bar{C} > 0 \end{cases}$$

$$\therefore \begin{bmatrix} P & X_{1,T} Q \\ Q X_{1,T}^T & P \end{bmatrix} > 0$$

$$\therefore \begin{bmatrix} X_{0,T-1} Q & X_{1,T} Q \\ Q X_{1,T}^T & X_{0,T-1} Q \end{bmatrix} > 0 \text{ when } K = U_{0,T-1} G_K$$

d)  $K$  can be computed, via selected  $G_K$ , such that

$$U_{0,T-1} G_K = K$$

subject to,  $X_{0,T-1} G_K = I$ , and other additional requirement such that stability and optimality.



**Exercise 2.** Considering discrete-time, infinite-horizon linear quadratic regulator problems, it can be shown that the minimiser  $K$  of the following semi-definite programme

$$\begin{aligned} \min_{K, P, L} \quad & \text{trace}(QP) + \text{trace}(L) \\ \text{subject to} \quad & \begin{cases} (A + BK)P(A + BK)^\top - P + I \leq 0 \\ P \geq I \\ L - R^{1/2}KPK^\top R^{1/2} \geq 0 \end{cases} \end{aligned}$$

coincides with the optimal gain  $K^*$

- The LQR problem is defined by a certain class of dynamics and cost functionals. Write expressions for the form of the dynamics and cost functionals are.
- Suppose that the dynamics are unknown. Define the data matrices  $U_{0,T-1}$ ,  $X_{0,T-1}$ ,  $X_{1,T}$ , as introduced in the handouts, which would (subject to a “suitable” condition) enable a data-driven solution of the LQR problem.
- State what the “suitable condition” in part (b) is. That is, state the condition on the collected data that ensures that the data is “sufficiently rich”.
- Show how, given that the condition in part (c) is satisfied, the Semi-definite programme given above can be replaced by the data-based semi-definite programme

$$\begin{aligned} \min_{\bar{Q}, L} \quad & \text{trace}(QX_{0,T-1}\bar{Q}) + \text{trace}(L) \\ \text{subject to} \quad & \begin{cases} X_{1,T}\bar{Q}(X_{0,T-1}\bar{Q})^{-1}\bar{Q}^\top X_{1,T}^\top - X_{0,T-1}\bar{Q} + I \leq 0 \\ X_{0,T-1}\bar{Q} \geq I \\ L - R^{1/2}U_{0,T-1}G_KPU_{0,T-1}G_K^\top R^{1/2} \geq 0 \\ X_{0,T-1}G_K = I. \end{cases} \end{aligned}$$

(a) dynamics :  $\dot{x} = Ax + Bu$  or  $x(k+1) = Ax(k) + Bu(k)$

cost :  $J = \frac{1}{2} \int_0^\infty (x^\top Q x + u^\top R u) dt$  or  $J = \frac{1}{2} \sum_{k=0}^\infty (x_k^\top Q x_k + u_k^\top R u_k)$

(b)  $U_{0,T-1} = [u_d(0), u_d(1), u_d(2), \dots, u_d(T-1)] \in \mathbb{R}^{m \times T}$

$X_{0,T-1} = [x_d(0), x_d(1), x_d(2), \dots, x_d(T-1)] \in \mathbb{R}^{n \times T}$

$X_{1,T} = [x_d(1), x_d(2), x_d(3), \dots, x_d(T)] \in \mathbb{R}^{n \times T}$

$$(c) \quad m + n \leq T$$

(d)

$$A + BK = [A \ B] \begin{bmatrix} I \\ K \end{bmatrix} = [A \ B] \left[ \frac{X_{0,T-1}}{U_{0,T-1}} \right] G_K = X_{1,T} G_K$$

$$\text{Let } Q > 0, \text{ where } Q = G_K P$$

$$\therefore X_{0,T-1} Q = X_{0,T-1} G_K P = I P = P$$

$$U_{0,T-1} Q = U_{0,T-1} G_K P = K P$$

$$\therefore P = G_K P P^{-1} = Q P^{-1}$$

$$\begin{aligned} \therefore (A + BK) P (A + BK)^T - P + I &= X_{1,T} G_K P G_K^T X_{1,T}^T - P + I \\ &= X_{1,T} Q P^{-1} (G_K P)^T X_{1,T}^T - P + I \\ &= X_{1,T} Q (X_{0,T-1} Q)^{-1} Q^T X_{1,T}^T - X_{0,T-1} Q + I \geq 0 \end{aligned}$$

$$P = X_{0,T-1} Q \geq I$$

$$L - R^T K P K^T R^T = L - R^T U_{0,T-1} G_K P G_K^T U_{0,T-1}^T R^T \geq 0$$

$$I = X_{0,T-1} G_K$$

**Exercise 3.** Consider a system

$$\dot{x} = Ax + Bu,$$

where  $x \in \mathbb{R}^2$  and  $u \in \mathbb{R}$  and suppose the matrices  $A$  and  $B$  are unknown.

(a) Consider the following data sets

$$\text{Data set 1: } x_d(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad x_d(1) = \begin{bmatrix} 3 \\ 0.5 \end{bmatrix}, \quad x_d(2) = \begin{bmatrix} 6.25 \\ 2 \end{bmatrix},$$

$$u_d(0) = 1, \quad u_d(1) = 0, \quad u_d(2) = -1.$$

$$\text{Data set 2: } x_d(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad x_d(1) = \begin{bmatrix} 3 \\ 0.5 \end{bmatrix}, \quad x_d(2) = \begin{bmatrix} 6.25 \\ 2 \end{bmatrix},$$

$$x_d(3) = \begin{bmatrix} 12.5 \\ 5.125 \end{bmatrix}$$

$$u_d(0) = 1, \quad u_d(1) = 0, \quad u_d(2) = -1.$$

$$\text{Data set 3: } x_d(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad x_d(1) = \begin{bmatrix} 3 \\ 0.5 \end{bmatrix}, \quad x_d(2) = \begin{bmatrix} 7.25 \\ 2 \end{bmatrix},$$

$$x_d(3) = \begin{bmatrix} 16.5 \\ 5.625 \end{bmatrix}$$

$$u_d(0) = 1, \quad u_d(1) = 1, \quad u_d(2) = 1.$$

Which of the above datasets, if any, are such that they can be used to obtain a purely data-based representation of the system? Explain your reasoning.

- (b) Using one of the data sets provided above, identify the system matrices  $A$  and  $B$ .
- (c) Without identifying the system, use one of the data sets provided above to design a stabilising controller. You may find it useful to use cvx to solve an appropriate (data-based) optimisation problem.

(a) Since for  $X_{0:T-1} \in \mathbb{R}^{n \times T}$ ,  $X_{1:T} \in \mathbb{R}^{n \times T}$  and  $U_{0:T-1} \in \mathbb{R}^{m \times T}$ , there must have

$T \geq n+m$  for sufficient rich data, which data set 2 and 3 both suitable

(b) for data set 2:

$$\begin{bmatrix} X_{0:T-1} \\ U_{0:T-1} \end{bmatrix} = \begin{bmatrix} 1 & 3 & 6.25 \\ 0 & 0.5 & 2 \\ 1 & 0 & -1 \end{bmatrix}, \quad X_{1:T} = \begin{bmatrix} 3 & 6.25 & 12.5 \\ 0.5 & 2 & 5.125 \end{bmatrix}$$

$$\therefore \begin{bmatrix} X_{0:T-1} \\ U_{0:T-1} \end{bmatrix}^+ = \begin{bmatrix} X_{0:T-1} \\ U_{0:T-1} \end{bmatrix}^T \left( \begin{bmatrix} X_{0:T-1} \\ U_{0:T-1} \end{bmatrix} \begin{bmatrix} X_{0:T-1} \\ U_{0:T-1} \end{bmatrix}^T \right)^{-1}$$

$$= \begin{bmatrix} 1 & 3 & 6.25 \\ 0 & 0.5 & 2 \\ 1 & 0 & -1 \end{bmatrix}^T \begin{bmatrix} 49.0625 & 14 & -5.25 \\ 14 & 4.25 & -2 \\ -5.25 & -2 & 2 \end{bmatrix}^{-1}$$

$$\begin{aligned}
&= \begin{bmatrix} 1 & 3 & 6.25 \\ 0 & 0.5 & 2 \\ 1 & 0 & -1 \end{bmatrix}^T \frac{1}{-5.25 \cdot -5.6875 + 2 \cdot -24.625 + 2 \cdot 12.515625} \begin{bmatrix} 4.5 & -17.5 & -5.6875 \\ -17.5 & 70.5625 & 24.625 \\ -5.6875 & 24.625 & 12.515625 \end{bmatrix}^T \\
&= \begin{bmatrix} 1 & 3 & 6.25 \\ 0 & 0.5 & 2 \\ 1 & 0 & -1 \end{bmatrix}^T \frac{64}{361} \begin{bmatrix} 4.5 & -17.5 & -5.6875 \\ -17.5 & 70.5625 & 24.625 \\ -5.6875 & 24.625 & 12.515625 \end{bmatrix} \\
&= \begin{bmatrix} -0.2105 & 1.2632 & 1.2105 \\ 0.8421 & -3.0526 & -0.8421 \\ -0.2105 & 1.2632 & 0.2105 \end{bmatrix}
\end{aligned}$$

$$\therefore [A \ B] = X_{1,T} \left[ \frac{X_{0,T-1}}{U_{0,T-1}} \right]^T = \begin{bmatrix} 2 & 0.5 & 1 \\ 0.5 & 1 & 0 \end{bmatrix}$$

$$\therefore A = \begin{bmatrix} 2 & 0.5 \\ 0.5 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Similarly, for data set 3:

$$A = \begin{bmatrix} 2 & 0.5 \\ 0.5 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

(c) for data set 2,

$$\begin{bmatrix} X_{0,T-1} Q & X_{1,T} Q \\ Q X_{1,T}^T & X_{0,T-1} Q \end{bmatrix} > 0 \quad \text{where } Q = G_K P$$

find Q via cvx

$$\therefore K = U_{0,T-1} G_K = U_{0,T-1} Q (X_{0,T-1} Q)^{-1} = [-1.3205 \quad -0.5]$$

$$\text{or } U_{0,T-1} G_K = U_{0,T-1} Q X_{0,T-1}^T (X_{0,T-1} Q X_{0,T-1}^T)^{-1} = [-1.4033 \quad -0.5655]$$